

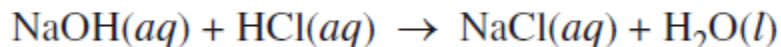
Chapter 11 Part 4: Titration

1.

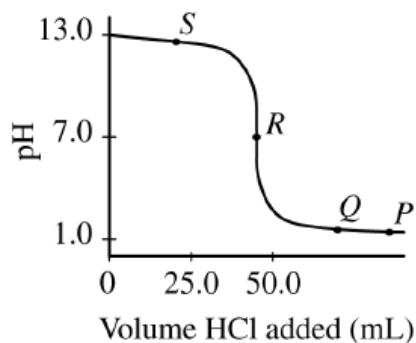
Initial buret reading	2.17 mL
Final buret reading	22.17 mL

An antacid tablet containing $\text{Mg}(\text{OH})_2(s)$ (molar mass 58.3 g/mol) is titrated with a 0.100 M solution of $\text{HNO}_3(aq)$. The end point is determined by using an indicator. Based on the data in the table above, what was the mass of the $\text{Mg}(\text{OH})_2(s)$ in the antacid tablet?

- (A) 0.0583 g
(B) 0.583 g
(C) 5.83 g
(D) 58.3 g



To determine the concentration of a $\text{NaOH}(aq)$ solution, a student titrated a 50. mL sample with 0.10 M $\text{HCl}(aq)$. The reaction is represented by the equation above. The titration is monitored using a pH meter, and the experimental results are plotted in the graph below.



2. A student mixes a 10.0 mL sample of 1.0 M $\text{NaOH}(aq)$ with a 10.0 mL sample of 1.0 M $\text{HCl}(aq)$ in a polystyrene container. The temperature of the solutions before mixing was 20.0°C. If the final temperature of the mixture is 26.0°C, what is the experimental value of $\Delta H_{\text{rxn}}^\circ$? (Assume that the solution mixture has a specific heat of 4.2 J/(g•K) and a density of 1.0 g/mL.)

- (A) -50. kJ/mol_{rxn}
(B) -25 kJ/mol_{rxn}
(C) -5.0×10^4 kJ/mol_{rxn}
(D) -5.0×10^2 kJ/mol_{rxn}

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3.

Trial	Volume of 0.10 M HCl	Volume of 0.10 M NaOH	Amount of Heat Released
1	50. mL	50. mL	X
2	100. mL	50. mL	Y

A student conducted an experiment to determine $\Delta H_{\text{rxn}}^{\circ}$ for the reaction between $\text{HCl}(aq)$ and $\text{NaOH}(aq)$.

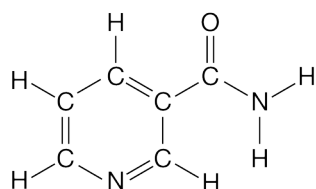
The student ran two trials using the volumes of $\text{HCl}(aq)$ and $\text{NaOH}(aq)$ indicated in the table above, and determined the amount of heat released. Which of the following best explains the relationship between X and Y?

- (A) $Y = 2X$, because the volume of $\text{HCl}(aq)$ used in trial 2 is twice the volume used in trial 1.
- (B) $Y = X$, because the number of moles of acid and base reacting with each other is the same in both trials.
- (C) $Y = 2X / 3$, because the heat is distributed over more particles in trial 2 than in trial 1.
- (D) The relationship between X and Y cannot be predicted.
4. At the point labeled R on the pH curve, which of the following ions are present in the reaction mixture at a concentration greater than 0.01 M?
- (A) Na^+ and Cl^- only
- (B) Na^+ , Cl^- and H^+ only
- (C) Na^+ , Cl^- and OH^- only
- (D) Na^+ , Cl^- , H^+ and OH^-
5. One student titrated the $\text{NaOH}(aq)$ with 1.0 M $\text{HCl}(aq)$ instead of 0.10 M $\text{HCl}(aq)$. How would the student's titration curve differ from the original curve?
- (A) The initial pH would be 11 instead of 13.
- (B) The pH at the equivalence point would be 5 instead of 7.
- (C) The pH far beyond the equivalence point would be higher than in the original curve.
- (D) The pH far beyond the equivalence point would be lower than in the original curve.
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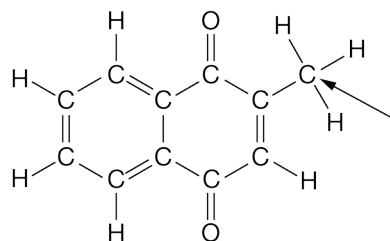
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6. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates properties of two compounds found in food, labeled Compound W and Compound Y. Structures of the compounds are shown below.



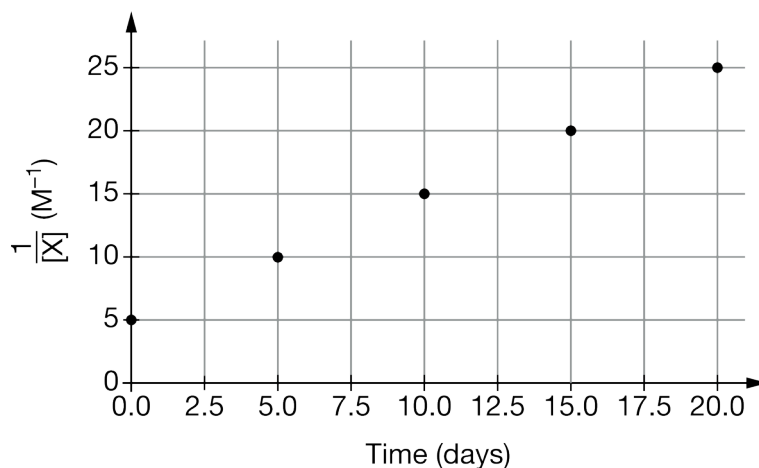
Compound W
(molar mass = 122 g/mol)



Compound Y
(molar mass = 172 g/mol)

- (a) Identify the hybridization of the valence orbitals of the carbon atom indicated with the arrow in the diagram of compound Y.
- (b) List all of the type(s) of intermolecular force(s) that the two compounds have in common.
- (c) Compound W has a greater boiling point than compound Y. The student claims that the difference is because compound Y exhibits stronger hydrogen bonding than compound W. Do you agree or disagree with this claim? Justify your answer in terms of the types and relative strengths of intermolecular forces in each substance.

A different compound found in food, compound X, is known to decompose over time. To study the reaction, a student monitors the molar concentration of compound X in a solution at constant temperature. After collecting and analyzing the data, the following linear graph was created by plotting $1/[X]$ versus time.



- (d) The student proposes that the reaction is second order with respect to compound X. Do you agree or disagree?

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Justify your answer using the data in the graph above.

(e) Write the rate law expression for the decomposition of compound **X** that is consistent with the graph above.

(f) Determine the value of the rate constant, k , for the reaction. Include units with your answer.

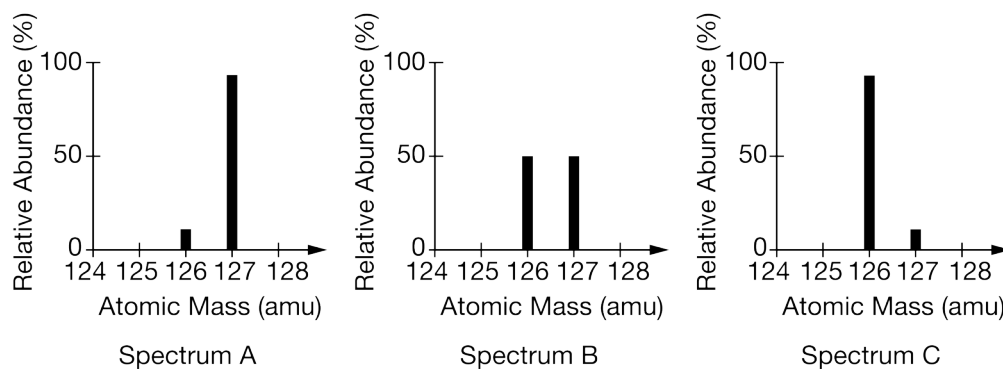
(g) The student claims that if the experiment were conducted again at a higher temperature, the value for the rate constant, k , would decrease. Do you agree or disagree with the student's claim? Justify your answer by comparing the slope of the original line above to the slope that would result from the higher temperature conditions.

A student titrates a solution of compound **X** to determine the concentration of **X** in the solution. In order to conduct the titration, the student needs to prepare 100. mL of 0.025 M $KIO_3(aq)$ by diluting a 0.500 M stock solution.

(h) What volume of 0.500 M stock solution, in mL, would be required to prepare 100. mL of 0.025 M $KIO_3(aq)$?

(i) To prepare the diluted solution, the student uses a graduated cylinder to measure the volume of stock solution calculated in part (h) and adds it to a 100.0 mL volumetric flask, followed by the addition of distilled water. If the student spilled some of the stock solution while transferring it to the volumetric flask and continues to fill it to the line with distilled water, would the resulting solution have a concentration that is greater than, less than, or equal to the intended concentration? Justify your answer.

The $IO_3^-(aq)$ ion can be reduced to form elemental iodine. Three possible mass spectra for a sample of atomic iodine (**I**) are represented below.



(j) Identify the mass spectrum that represents the isotopes of iodine. Justify your choice.

Iodine can react with a certain element, **M**. The successive ionization energy values for element **M** are given below.

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	Ionization Energy (kJ/mol)
First	578
Second	1817
Third	2745
Fourth	11,578
Fifth	14,842

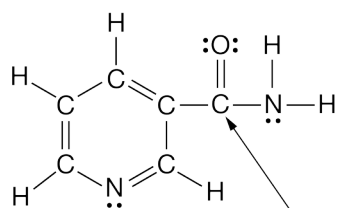
(k) What is the charge of the ion formed by element **M** in its compounds? Justify your answer using the data in the table.

(l) Given the charge determined in part (k), what is the formula for the compound that element **M** would form with iodine?

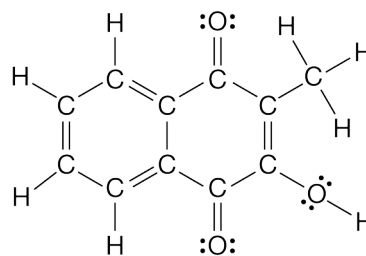
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7. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates properties of two compounds found in food, labeled Compound W and Compound Z. Structures of the compounds are shown below.



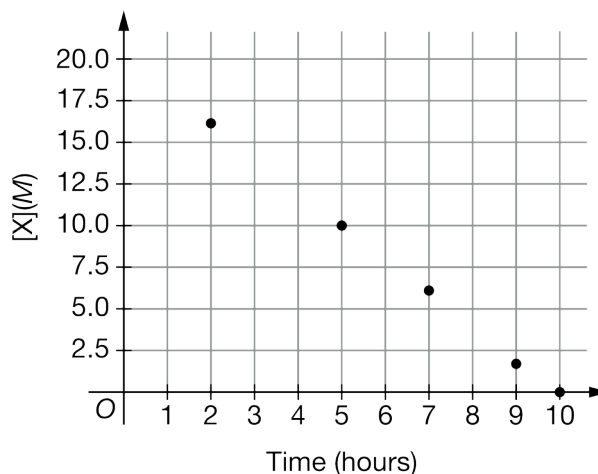
Compound W
(molar mass = 122 g/mol)



Compound Z
(molar mass = 186 g/mol)

- (a) Using VSEPR theory, predict the O — C — N bond angle indicated by the arrow in the diagram of compound W.
- (b) List all of the type(s) of intermolecular force(s) that the two compounds have in common.
- (c) Compound Z has a higher boiling point than compound W. The student claims that compound Z has a greater boiling point because it has greater London dispersion forces than compound W. Do you agree or disagree with this claim? Justify your answer in terms of the types and relative strengths of intermolecular forces in each substance.

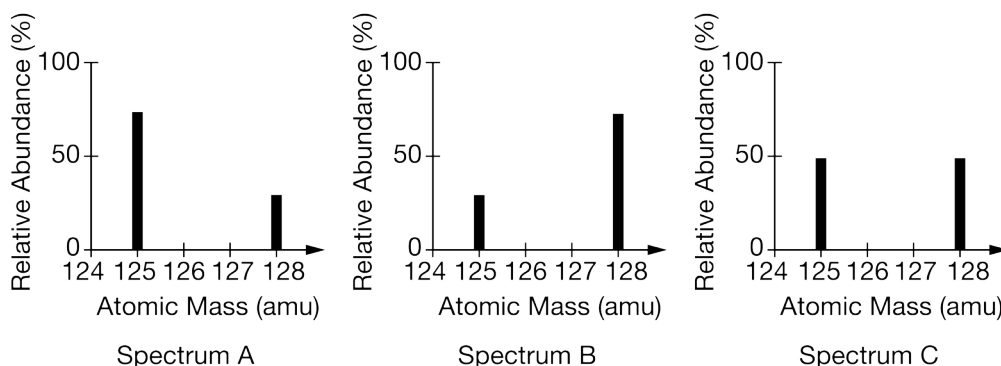
A different compound found in food, compound X is known to decompose over time. To study the reaction, a student monitors the molar concentration of compound X in a solution at constant temperature. After collecting and analyzing the data, the following linear graph was created by plotting [X] versus time.



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- (d) The student proposes that the reaction is zero order with respect to compound X. Do you agree or disagree? Justify your answer using the data in the graph above.
- (e) Write the rate law expression for the decomposition of compound X that is consistent with the graph above.
- (f) Determine the value of the rate constant, k , for the reaction. Include units with your answer.
- (g) The student claims that if the experiment were conducted again at a higher temperature, the value for the rate constant, k , would increase. Do you agree or disagree with the student's claim? Justify your answer by comparing the slope of the original line above to the slope that would result from the higher temperature conditions.
- A student titrates a solution of compound X to determine the concentration of X in the solution. In order to conduct the titration, the student needs to prepare 100. mL of 0.010 M $KIO_3(aq)$ by diluting a 0.040 M stock solution.
- (h) What volume of 0.040 M stock solution, in mL, would be required to prepare 100. mL of 0.010 M $KIO_3(aq)$?
- (i) To prepare the diluted solution, the student uses a graduated cylinder to measure the volume of stock solution calculated in part (h) and adds it to a 100.0 mL volumetric flask, followed by the addition of distilled water. If the student filled the volumetric flask with distilled water but the bottom of the meniscus was significantly above the 100.0 mL mark, would the resulting solution concentration be greater than, less than, or equal to what was intended? Justify your answer.

The $IO_3^-(aq)$ ion can be reduced to form elemental iodine. Three possible mass spectra for a sample of atomic iodine (I) are represented below.



- (j) Identify the mass spectrum that most likely represents the isotopes of iodine. Justify your choice.

Iodine can also react with a certain element, M. The successive ionization energy values for element M are given below.

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	Ionization Energy (kJ/mol)
First	419
Second	3051
Third	4419
Fourth	5878
Fifth	7975

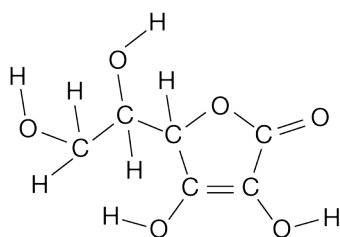
(k) What is the charge of the ion formed by element **M** in its compounds? Justify your answer using the data in the table.

(l) Given the charge determined in part (k), what is the formula for the compound that element **M** would form with iodine?

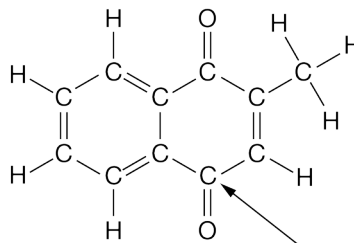
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8. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates properties of two compounds found in food, labeled Compound X and Compound Y. Structures of the compounds are shown below.



Compound X
(molar mass = 176 g/mol)

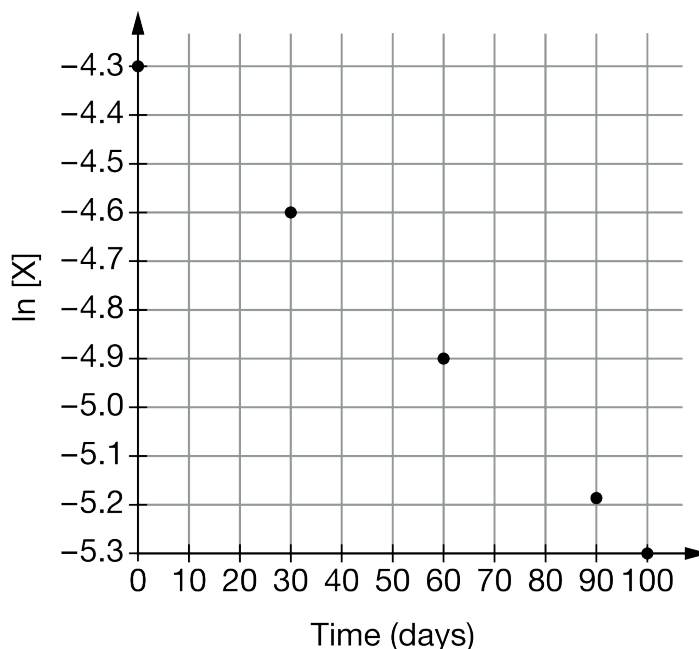


Compound Y
(molar mass = 172 g/mol)

- (a) Identify the hybridization of the valence orbitals of the carbon atom indicated with the arrow in the diagram of compound Y.
- (b) List all of the type(s) of intermolecular force(s) that the two compounds have in common.
- (c) Compound X has a greater boiling point than compound Y. The student claims that the difference is because compound X exhibits hydrogen bonding, but compound Y does not. Do you agree or disagree with this claim? Justify your answer in terms of the types and relative strengths of intermolecular forces in each substance.

To show that compound X present in food is lost over time, a student monitors the molar concentration of compound X in a solution at constant temperature. After collecting and analyzing the data, the following linear graph was created by plotting $\ln [X]$ versus time.

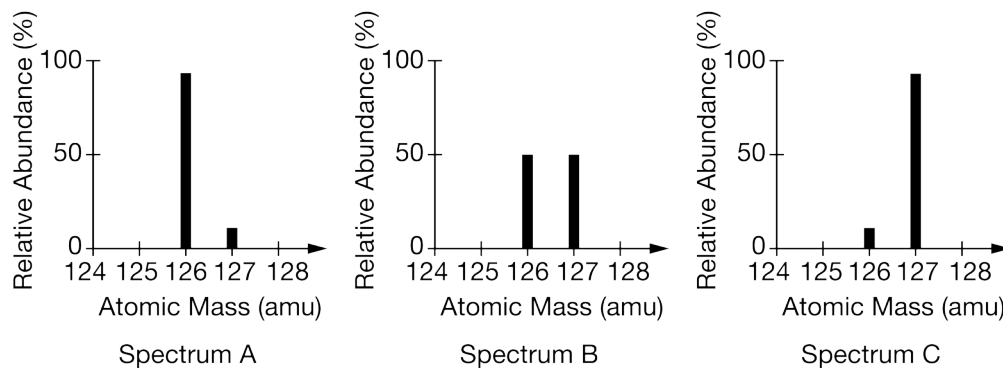
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- (d) The student proposes that the reaction is first order with respect to compound X. Do you agree or disagree? Justify your answer using the data in the graph above.
- (e) Write the rate law expression for the decomposition of compound X that is consistent with the graph above.
- (f) Determine the value of the rate constant, k , for the reaction. Include units with your answer.
- (g) The student claims that if the experiment were conducted again at a lower temperature, the value for the rate constant, k , would decrease. Do you agree or disagree with the student's claim? Justify your answer by comparing the slope of the original line above to the slope that would result from the lower temperature conditions.
- A student titrates a solution of compound X to determine the concentration of X in the solution. In order to conduct the titration, the student needs to prepare 100. mL of 0.050 M $\text{KIO}_3(aq)$ by diluting a 0.500 M stock solution.
- (h) What volume of 0.500 M stock solution, in mL, would be required to prepare 100. mL of 0.050 M $\text{KIO}_3(aq)$?
- (i) To prepare the diluted solution, the student uses a graduated cylinder to measure the volume of stock solution calculated in part (h) and adds it to a 100.0 mL volumetric flask, followed by the addition of distilled water. If the student fills the volumetric flask with distilled water, but the bottom of the meniscus is significantly below the 100.0 mL mark, would the resulting solution concentration be greater than, less than, or equal to the intended concentration? Justify your answer.

The $\text{IO}_3^- (aq)$ ion can be reduced to form elemental iodine. Three possible mass spectra for a sample of atomic iodine (I) are represented below.

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(j) Identify the mass spectrum that represents the isotopes of iodine. Justify your choice.

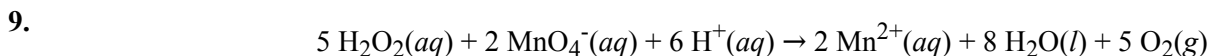
Iodine can react with a certain element, M . The successive ionization energy values for element M are given below.

	Ionization Energy (kJ/mol)
First	738
Second	1451
Third	7733
Fourth	10,542
Fifth	13,636

(k) What is the charge of the ion formed by element M in its compounds? Justify your answer using the data in the table.

(l) Given the charge determined in part (k), what is the formula for the compound that element M would form with iodine?

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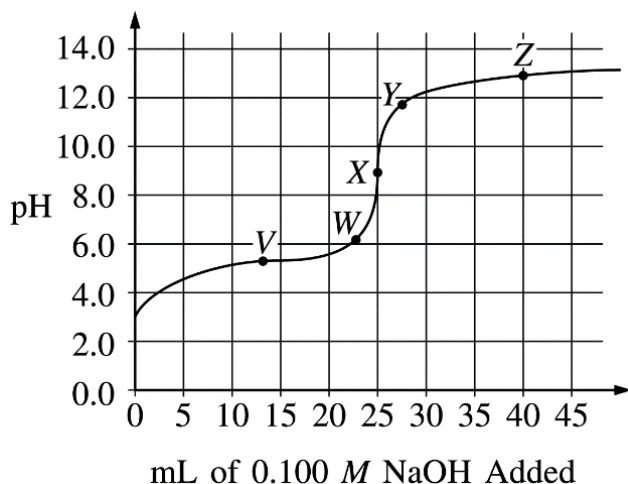


In a titration experiment, $\text{H}_2\text{O}_2(aq)$ reacts with aqueous $\text{MnO}_4^-(aq)$ as represented by the equation above. The dark purple KMnO_4 solution is added from a buret to a colorless, acidified solution of $\text{H}_2\text{O}_2(aq)$ in an Erlenmeyer flask. (Note: At the end point of the titration, the solution is a pale pink color.)

Which of the following best describes what happens to the pH of the H_2O_2 solution as the titration proceeds?

- (A) The +2 charge on the manganese ions maintains the acidity of the solution.
- (B) The production of water dilutes the solution, making it basic.
- (C) As H^+ ions are consumed, the solution becomes less acidic and the pH increases.
- (D) As H^+ ions are consumed, the solution becomes less acidic and the pH decreases.

The graph below shows the titration curve that results when 100. mL of 0.0250 *M* acetic acid is titrated with 0.100 *M* NaOH.



10. Which of the following indicators is the best choice for this titration?

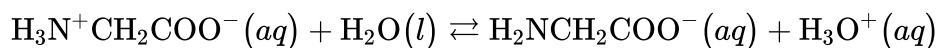
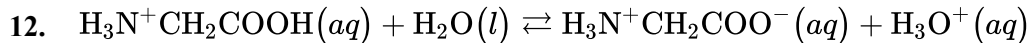
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(A)	<u>Indicator</u>	<u>pH Range of Color Change</u>
	Methyl orange	3.2 - 4.4
(B)	<u>Indicator</u>	<u>pH Range of Color Change</u>
	Methyl red	3.2 - 4.4
(C)	<u>Indicator</u>	<u>pH Range of Color Change</u>
	Bromothymol	6.1 - 7.6
(D)	<u>Indicator</u>	<u>pH Range of Color Change</u>
	Phenolphthalein	8.2 - 10.0
(E)	<u>Indicator</u>	<u>pH Range of Color Change</u>
	Alizarin	11.0 - 12.4

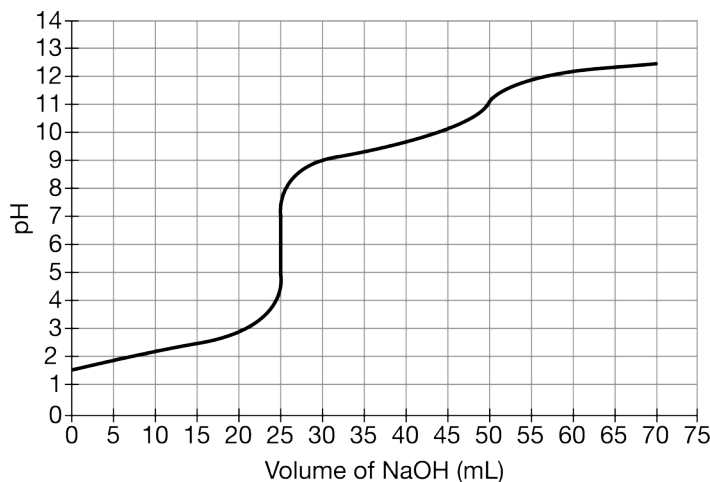
11. What part of the curve corresponds to the optimum buffer action for the acetic acid/acetate ion pair?

- (A) Point *V*
 - (B) Point *X*
 - (C) Point *Z*
 - (D) Along all of section *WY*
 - (E) Along all of section *YZ*
-

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The stepwise dissociation of the amino acid glycine is represented by the chemical equations above. A student titrates a sample of glycine dissolved in dilute acid with $0.100\text{ M NaOH}(aq)$. The data are plotted on the following graph.

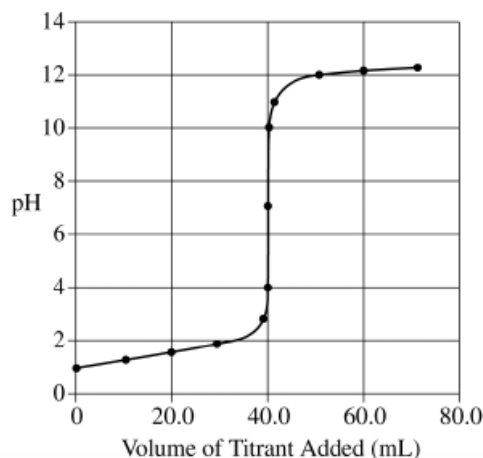


Based on the data, which of the following species has the highest concentration in an aqueous solution of glycine with a pH of 7?

- (A) $\text{H}_3\text{N}^+\text{CH}_2\text{COOH}(aq)$
- (B) $\text{H}_3\text{N}^+\text{CH}_2\text{COO}^-(aq)$
- (C) $\text{H}_2\text{NCH}_2\text{COO}^-(aq)$
- (D) $\text{H}_2\text{NCH}_2\text{COOH}(aq)$

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13.



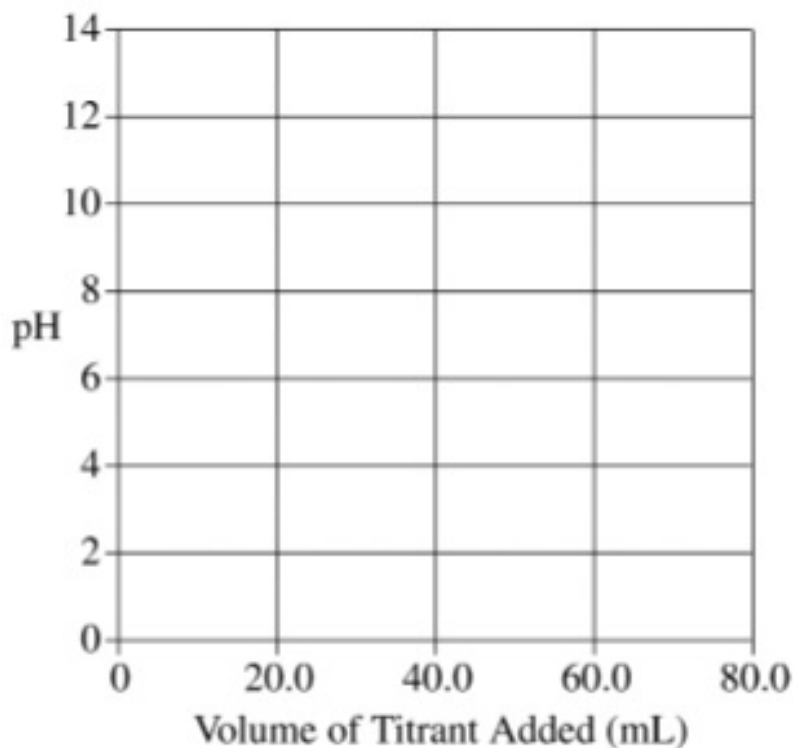
A solution of 0.100 M HCl and a solution of 0.100 M NaOH are prepared. A 40.0 mL sample of one of the solutions is added to a beaker and then titrated with the other solution. A pH electrode is used to obtain the data that are plotted in the titration curve shown above.

- Identify the solution that was initially added to the beaker. Explain your reasoning.
- On the titration curve above, circle the point that corresponds to the equivalence point.
- At the equivalence point, how many moles of titrant have been added?
- The same titration is to be performed again, this time using an indicator. Use the information in the table below to select the best indicator for the titration. Explain your choice.

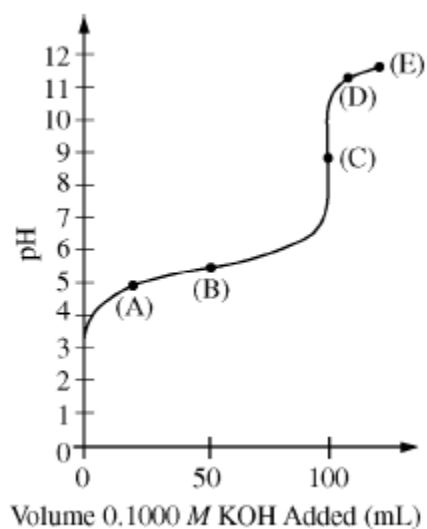
Indicator	pH Range of Color Change
Methyl violet	0 – 1.6
Methyl red	4 – 6
Alizarin yellow	10 – 12

- What is the difference between the equivalence point of a titration and the end point of a titration?
- On the grid provided on the next page, sketch the titration curve that would result if the solutions in the beaker and buret were reversed (i.e., if 40.0 mL of the solution used in the buret in the previous titration were titrated with the solution that was in the beaker).

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14. A solution of a weak monoprotic acid is titrated with a solution of a strong base, KOH. Consider the points labeled (A) through (E) on the titration curve that results, as shown below.

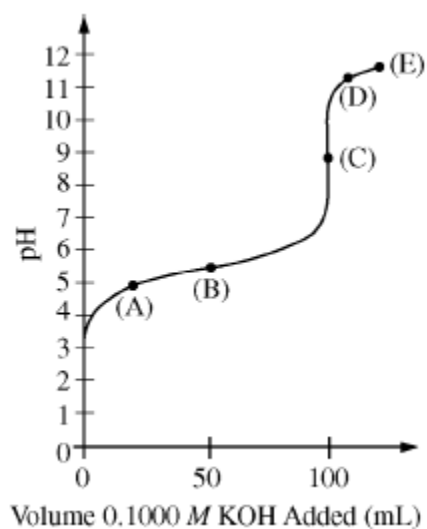


The point at which the moles of the added strong base are equal to the moles of the weak acid initially present

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- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

15. A solution of a weak monoprotic acid is titrated with a solution of a strong base, KOH. Consider the points labeled (A) through (E) on the titration curve that results, as shown below.

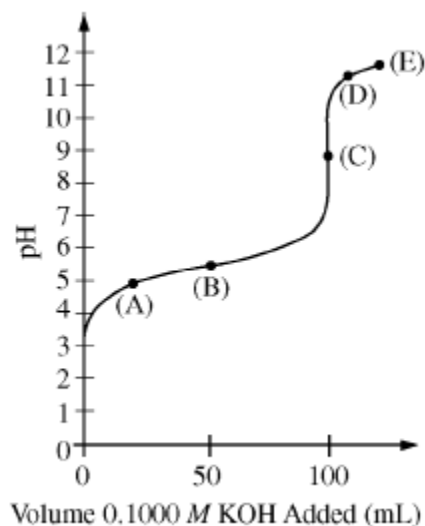


The point at which the pH is closest to that of the strong base being added

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

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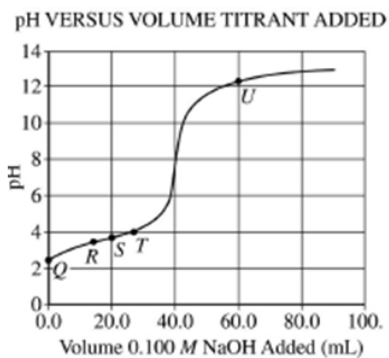
16. A solution of a weak monoprotic acid is titrated with a solution of a strong base, KOH. Consider the points labeled (A) through (E) on the titration curve that results, as shown below.



The point at which the concentrations of the weak acid and its conjugate base are approximately equal

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

Refer to the following.



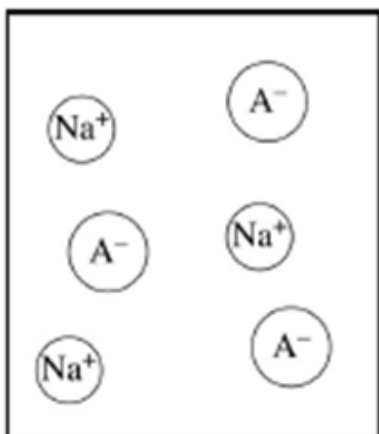
A 50.0 mL sample of an acid, HA, of unknown molarity is titrated, and the pH of the resulting solution is measured with a meter and graphed as a function of the volume of 0.100 M NaOH added.

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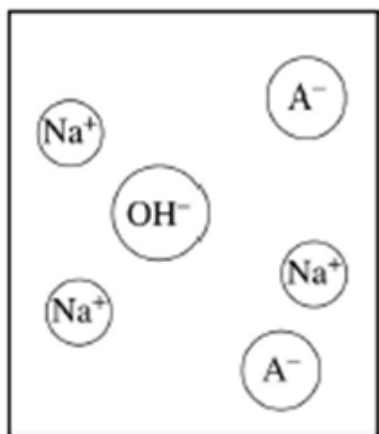
17. A student carries out the same titration but uses an indicator instead of a pH meter. If the indicator changes color slightly past the equivalence point, what will the student obtain for the calculated concentration of the acid?
- (A) Slightly less than 0.0800 *M*
 - (B) Slightly more than 0.0800 *M*
 - (C) Slightly less than 0.125 *M*
 - (D) Slightly more than 0.125 *M*
18. At point *R* in the titration, which of the following species has the highest concentration?
- (A) HA
 - (B) A[−]
 - (C) H₃O⁺
 - (D) OH[−]
19. At which point on the titration curve is [A[−]] closest to twice that of [HA]?
- (A) *R*
 - (B) *S*
 - (C) *T*
 - (D) *U*
20. Which of the following is the best particulate representation of the species (other than H₂O) that are present in significant concentrations in the solution at point *U* in the titration?

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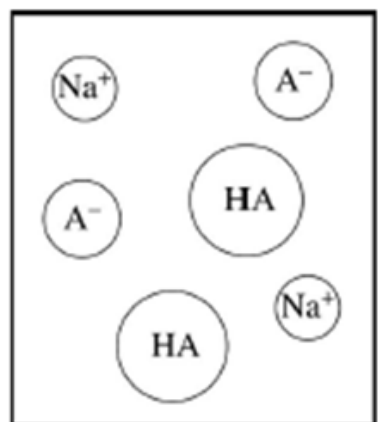
(A)



(B)

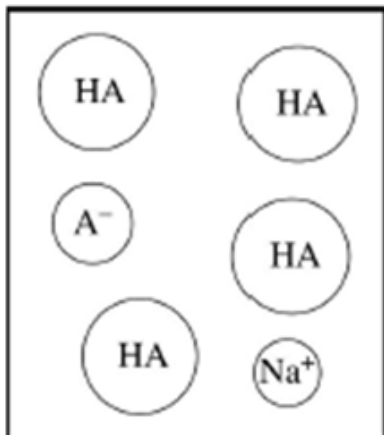


(C)

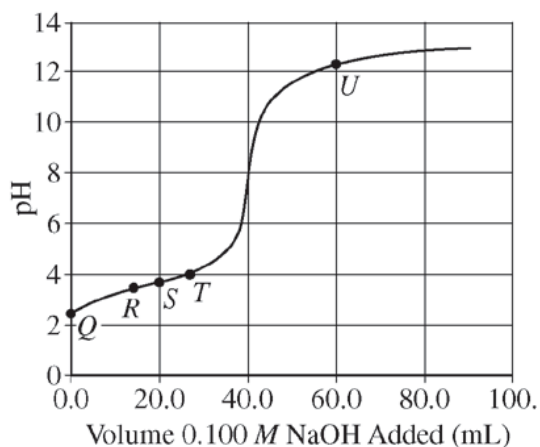


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(D)



pH VERSUS VOLUME TITRANT ADDED



A 50.0 mL sample of an acid, HA, of unknown molarity is titrated, and the pH of the resulting solution is measured with a pH meter and graphed as a function of the volume of 0.100 M NaOH added.

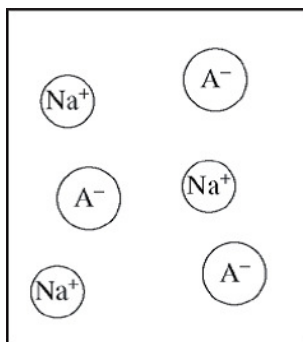
21. A student carries out the same titration, but uses an indicator instead of a pH meter. If the indicator changes color slightly past the equivalence point, what will the student obtain for the calculated concentration of the acid?
- (A) Slightly less than 0.0800 M
 - (B) Slightly more than 0.0800 M
 - (C) Slightly less than 0.125 M
 - (D) Slightly more than 0.125 M
22. At point R in the titration, which of the following species has the highest concentration?
- (A) HA
 - (B) A⁻
 - (C) H₃O⁺
 - (D) OH⁻

Chapter 11 Part 4: Titration

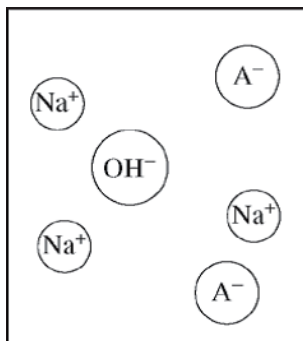
23. At which point on the titration curve is $[A^-]$ closest to twice that of $[HA]$?
- (A) *R*
 - (B) *S*
 - (C) *T*
 - (D) *U*
24. Which of the following is the best particulate representation of the species (other than H_2O) that are present in significant concentrations in the solution at point *U* in the titration?

Chapter 11 Part 4: Titration

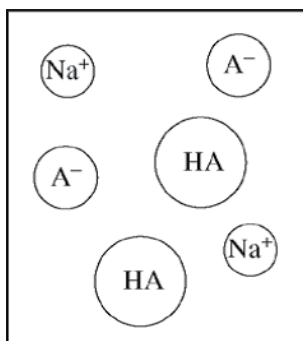
(A)



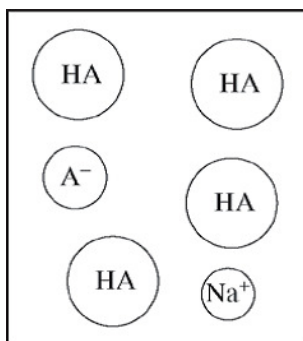
(B)



(C)



(D)



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25. A student pipetted five 25.00-milliliter samples of hydrochloric acid and transferred each sample to an Erlenmeyer flask, diluted it with distilled water, and added a few drops of phenolphthalein to each. Each sample was then titrated with a sodium hydroxide solution to the appearance of the first permanent faint pink color. The following results were obtained.

Volumes of NaOH Solution

First Sample.....35.22 mL

Second Sample.....36.14 mL

Third Sample.....36.13 mL

Fourth Sample36.15 mL

Fifth Sample.....36.12 mL

Which of the following is the most probable explanation for the variation in the student's results?

- (A) The burette was not rinsed with NaOH solution.
- (B) The student misread a 5 for a 6 on the burette when the first sample was titrated.
- (C) A different amount of water was added to the first sample.
- (D) The pipette was not rinsed with the HCl solution.
- (E) The student added too little indicator to the first sample.

26. $\text{C}_9\text{H}_8\text{O}_4(aq) + \text{NaOH}(aq) \rightleftharpoons \text{C}_9\text{H}_7\text{O}_4^-(aq) + \text{Na}^+(aq) + \text{H}_2\text{O}(l)$

The reaction represented above occurs when 2.00×10^{-4} mol of pure acetylsalicylic acid, $\text{C}_9\text{H}_8\text{O}_4$, is completely dissolved in 15.00 mL of water in a flask and titrated to the equivalence point with 0.100 M NaOH(aq). Which of the following statements about the titration is true at the equivalence point?

- (A) $[\text{C}_9\text{H}_8\text{O}_4]$ is greater than $[\text{C}_9\text{H}_7\text{O}_4^-]$.
- (B) $[\text{C}_9\text{H}_8\text{O}_4]$ is less than $[\text{C}_9\text{H}_7\text{O}_4^-]$.
- (C) $[\text{C}_9\text{H}_8\text{O}_4]$ is equal to $[\text{C}_9\text{H}_7\text{O}_4^-]$.
- (D) The relationship between $[\text{C}_9\text{H}_8\text{O}_4]$ and $[\text{C}_9\text{H}_7\text{O}_4^-]$ cannot be determined without additional information.

Chapter 11 Part 4: Titration

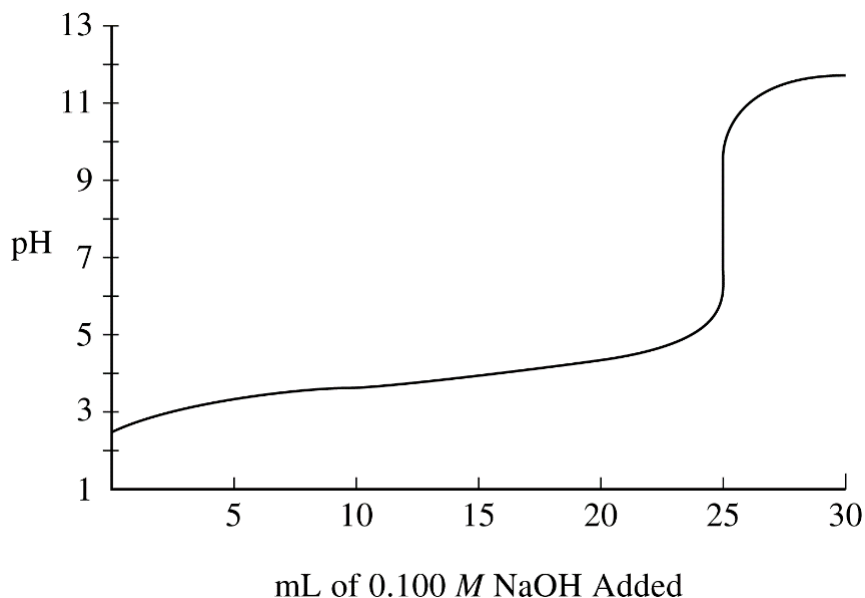
27.

Acid Solution	Volume of NaOH Added (mL)
A	40
B	75
C	115
D	200

To maximize the yield in a certain manufacturing process, a solution of a weak monoprotic acid that has a concentration between 0.20 M and 0.30 M is required. Four 100. mL samples of the acid at different concentrations are each titrated with a 0.20 M NaOH solution. The volume of NaOH needed to reach the end point for each sample is given in the table above. Which solution is the most suitable to maximize the yield?

- (A) Solution A
- (B) Solution B
- (C) Solution C
- (D) Solution D

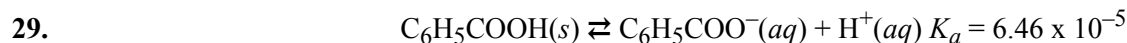
28.



An unknown acid is dissolved in 25 mL of water and titrated with 0.100 M NaOH. The results are shown in the titration curve above. Which of the following could be the unknown acid?

Chapter 11 Part 4: Titration

- (A) Fluoroacetic acid, $pK_a = 2.6$
- (B) Glycolic acid, $pK_a = 3.8$
- (C) Propanoic acid, $pK_a = 4.9$
- (D) Hypochlorous acid, $pK_a = 7.5$
- (E) Boric acid, $pK_a = 9.3$



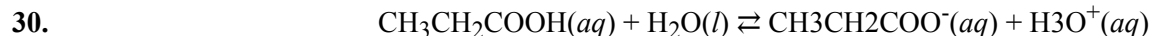
Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, dissociates in water as shown in the equation above. A 25.0 mL sample of an aqueous solution of pure benzoic acid is titrated using standardized 0.150 M NaOH.

- a. After addition of 15.0 mL of the 0.150 M NaOH, the pH of the resulting solution is 4.37. Calculate each of the following.
 - i. $[\text{H}^+]$ in the solution
 - ii. $[\text{OH}^-]$ in the solution
 - iii. The number of moles of NaOH added
 - iv. The number of moles of $\text{C}_6\text{H}_5\text{COO}^- (aq)$ in the solution
 - v. The number of moles of $\text{C}_6\text{H}_5\text{COOH}$ in the solution
- b. State whether the solution at the equivalence point of the titration is acidic, basic, or neutral. Explain your reasoning.

In a different titration, a 0.7529 g sample of a mixture of solid $\text{C}_6\text{H}_5\text{COOH}$ and solid NaCl is dissolved in water and titrated with 0.150 M NaOH. The equivalence point is reached when 24.78 mL of the base solution is added.

- c. Calculate each of the following.
 - i. The mass, in grams, of benzoic acid in the solid sample
 - ii. The mass percentage of benzoic acid in the solid sample

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Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, is a carboxylic acid that reacts with water according to the equation above. At 25°C the pH of a 50.0 mL sample of 0.20 M $\text{CH}_3\text{CH}_2\text{COOH}$ is 2.79.

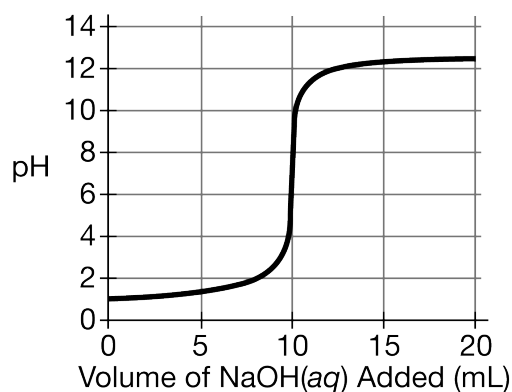
- Identify a Brønsted-Lowry conjugate acid-base pair in the reaction. Clearly label which is the acid and which is the base.
- Determine the value of K_a for propanoic acid at 25°C .
- For each of the following statements, determine whether the statement is true or false. In each case, explain the reasoning that supports your answer.
 - The pH of a solution prepared by mixing the 50.0 mL sample of 0.20 M $\text{CH}_3\text{CH}_2\text{COOH}$ with a sample 50.0 mL 0.20 M NaOH is 7.00.
 - If the pH of a hydrochloric acid solution is the same as the pH of a propanoic acid solution, then the molar concentration of the hydrochloric acid solution must be less than the molar concentration of the propanoic acid solution.

A student is given the task of determining the concentration of a propanoic acid solution of unknown concentration. A 0.173 M NaOH solution is available to use as the titrant. The student uses a 25.00 mL volumetric pipet to deliver the propanoic acid solution to a clean, dry flask. After adding an appropriate indicator to the flask, the student titrates the solution with the 0.173 M NaOH , reaching the end point after 20.52 mL of the base solution has been added.

- Calculate the molarity of the propanoic acid solution.
- The student is asked to redesign the experiment to determine the concentration of a butanoic acid solution instead of a propanoic acid solution. For butanoic acid the value of $\text{p}K_a$ is 4.83. The student claims that a different indicator will be required to determine the equivalence point of the titration accurately. Based on your response to part (b), do you agree with the student's claim? Justify your answer.

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31.



A 20.0 mL sample of a 0.125 M monoprotic acid solution is titrated with a $\text{NaOH}(aq)$ solution of unknown concentration. Based on the titration curve above, what is the molar concentration of the $\text{NaOH}(aq)$ solution?

- (A) 0.063 M
- (B) 0.125 M
- (C) 0.250 M
- (D) 0.500 M

32.

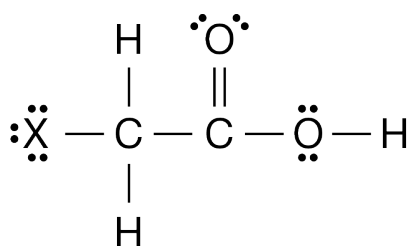
Acid	Concentration of Acid (M)	Volume of Sample (mL)	Volume of Standardized $\text{NaOH}(aq)$ Added to Reach Equivalence Point (mL)
$\text{HClO}_4(aq)$	0.100	25.0	24.0
$\text{HClO}(aq)$	0.200	25.0	?

Samples of two different acids are titrated using the same standardized solution of $\text{NaOH}(aq)$. According to the data in the table above, how many milliliters of the standardized $\text{NaOH}(aq)$ are required to reach the equivalence point when the 25.0 mL sample of 0.200 M $\text{HClO}(aq)$ is titrated?

- (A) 12.0 mL
- (B) 24.0 mL
- (C) 48.0 mL
- (D) It cannot be determined without knowing the values of K_a for $\text{HClO}_4(aq)$ and $\text{HClO}(aq)$.

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33. Potassium hydrogen phthalate, KHP, is used as a primary standard for determining the concentration of a solution of NaOH by titration. If the KHP has not been dried before weighing, the calculated molarity of the NaOH would be
- (A) higher than the actual value, since water is included in the apparent mass of KHP
 - (B) higher than the actual value, since the presence of water requires a larger volume of titrant
 - (C) lower than the actual value, since NaOH absorbs water
 - (D) unaffected, since KHP is a strong acid
 - (E) unaffected, since water is routinely added before the titration



The structure of haloacetic acids, XCH_2COOH (where X is either F, Cl, Br, or I), is shown above. The dissociation constants and molar masses of four haloacetic acids are listed in the table below.

Acid	$\text{p}K_a$	K_a	Molar Mass(g/mol)
Fluoroacetic acid	2.59	2.57×10^{-3}	78.0
Chloroacetic acid	2.87	1.35×10^{-3}	94.5
Bromoacetic acid	2.90	1.26×10^{-3}	138.9
Iodoacetic acid	3.18	6.61×10^{-4}	185.9

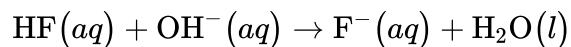
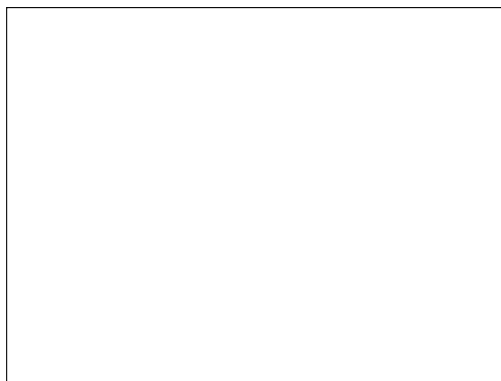
34. A student titrates 10.0 mL samples of 1.0 M solutions of each of the haloacetic acids with a standard solution of NaOH. Which of the following statements correctly predicts the volume of NaOH(aq) needed to reach the equivalence point?
- (A) Fluoroacetic acid will need the smallest volume of NaOH(aq) to reach the equivalence point.
 - (B) Iodoacetic acid will need the smallest volume of NaOH(aq) to reach the equivalence point.
 - (C) All of the acids will need the same volume of NaOH(aq) to reach the equivalence point.
 - (D) All of the haloacetic acids are weak; therefore none will reach an equivalence point.

Chapter 11 Part 4: Titration

35. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to HF and the titration of $\text{HF}(aq)$.

- (a) In the following box, draw the complete Lewis electron-dot diagram of the HF molecule.

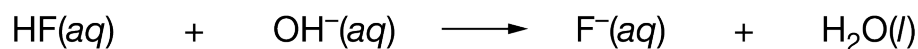
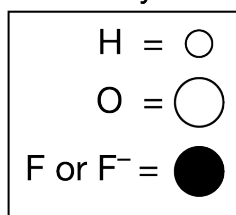


A student performed a titration of $\text{HF}(aq)$ with $\text{NaOH}(aq)$. The net ionic equation for the neutralization reaction that occurs during the titration is shown above.

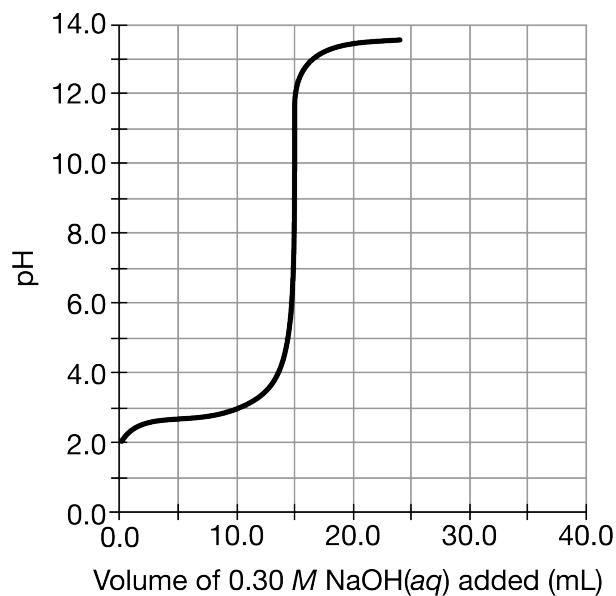
- (b) Identify both of the Brønsted-Lowry conjugate acid-base pairs in the neutralization reaction above. For each pair, label the acid and the base.
- (c) In terms of the species in the reaction, explain what occurs during the reaction that identifies it as an acid-base reaction.
- (d) The equation can be written in particulate form. Using the following key, draw the correct products above their formulas.

Chapter 11 Part 4: Titration

Key

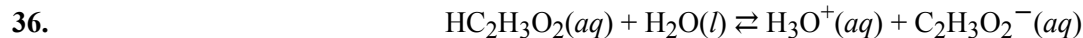


A student titrated 30. mL of 0.15 *M* HF(*aq*) with 0.30 *M* NaOH(*aq*). The results are plotted on the following graph.



- (e) How many moles of HF are in 30. mL of 0.15 *M* HF(*aq*)?
- (f) Draw an **X** on the horizontal axis of the graph to show the amount of NaOH(*aq*) that had been added when the equivalence point was reached.
- (g) The graph above shows the results of the titration of the 30. mL of 0.15 *M* HF(*aq*) with 0.30 *M* NaOH(*aq*). On the same graph, draw the curve for the titration of 60. mL of 0.15 *M* HF(*aq*) with 0.30 *M* NaOH(*aq*).

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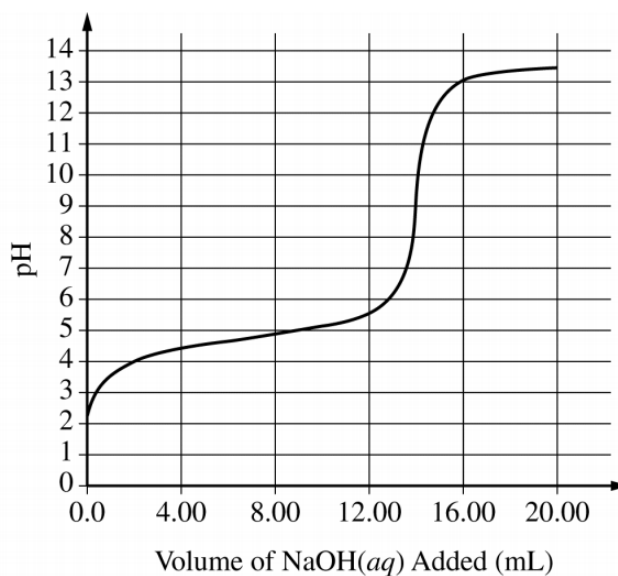
The dissociation of ethanoic acid, $\text{HC}_2\text{H}_3\text{O}_2(aq)$, is represented above. A student is given the task of determining the value of K_a for $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using two different experimental procedures.

- a. The student is first asked to prepare 100.0 mL of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using a 2.000 M standard solution.
- Calculate the volume, in mL, of 2.000 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ the student needs to prepare 100.0 mL of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$.
 - Describe the procedure the student should use to prepare 100. mL of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using appropriate equipment selected from the list below. Assume that the student uses appropriate safety equipment.
 - 100 mL beaker
 - 100 mL graduated cylinder
 - 100 mL volumetric flask
 - Eye dropper
 - 500 mL wash bottle filled with distilled water
 - 2.000 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ in a 50 mL buret
- b. Using a pH probe, the student determines that the pH of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ is 2.92.
- Using the pH value, calculate the value of K_a for $\text{HC}_2\text{H}_3\text{O}_2(aq)$
 - Calculate the percent dissociation of ethanoic acid in 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$

In a separate experimental procedure, the student titrates 10.0 mL of the 2.000 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ with an $\text{NaOH}(aq)$ solution of unknown concentration. The student monitors the pH during the titration. The following titration curve was created using the experimental data presented in the table.

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Volume of NaOH(aq) Added (mL)	pH
0.00	2.23
2.00	3.99
4.00	4.37
6.00	4.65
8.00	4.90
10.00	5.17
12.00	5.55
14.00	9.35
16.00	13.04
18.00	13.31
20.00	13.46



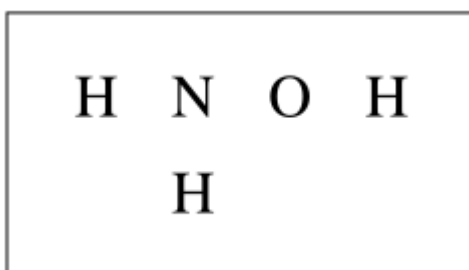
- c. Write the balanced net ionic equation for the reaction that occurs when $\text{HC}_2\text{H}_3\text{O}_2(aq)$ and $\text{NaOH}(aq)$ are combined.
- d. Calculate the molar concentration of the $\text{NaOH}(aq)$ solution.
- e. Explain how the student can estimate the value of K_a for $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using the titration curve.

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37. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about hydroxylamine, NH_2OH .

- (a) Complete the Lewis electron-dot diagram of the NH_2OH molecule in the box below, including all bonding and nonbonding valence electrons.

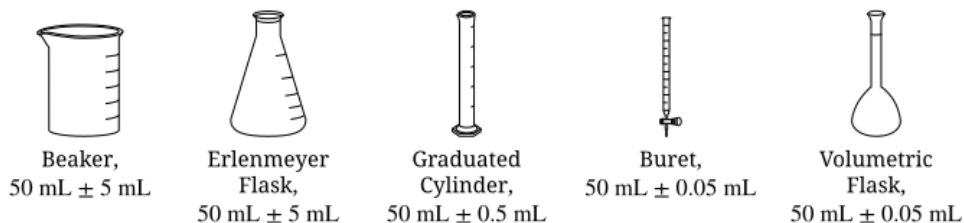


- (b) Hydroxylamine, $\text{NH}_2\text{OH}(aq)$, is a weak base. Write the balanced net ionic equation for the reaction that occurs when $\text{NH}_2\text{OH}(aq)$ and $\text{HCl}(aq)$ are combined.

A student is preparing to titrate a $\text{NH}_2\text{OH}(aq)$ solution with $\text{HCl}(aq)$. To prepare for the titration, the student places the NH_2OH solution into an Erlenmeyer flask.

- (c) The titration procedure instructs the student to prepare 100.00mL of a 0.1000M HCl solution. The stock solution available is 0.5604M HCl . Calculate the volume, in liters, of stock solution required for the dilution.

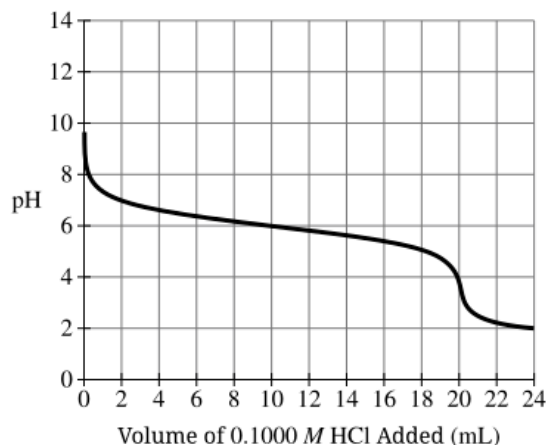
- (d) Using the following choices, identify the glassware that should be used to deliver the $\text{HCl}(aq)$ with the greatest precision. Circle your choice and provide a brief justification.



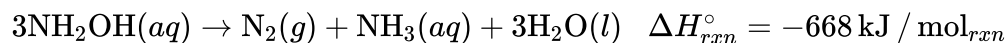
- (e) The student titrates a 25.00mL sample of a NH_2OH solution with 0.1000M HCl . The student monitors the pH during the experiment and creates the following titration curve. Selected data are given in the table.

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Volume of HCl Added (mL)	pH
0.00	9.67
2.00	7.00
6.00	6.40
8.00	6.21
10.00	6.03
12.00	5.85
14.00	5.66
18.00	5.08
20.00	3.44
24.00	2.01



- (i) Based on the data from the titration experiment, estimate the value of pK_b for NH_2OH .
- (ii) Calculate the molar concentration of the 25.00 mL sample of $\text{NH}_2\text{OH}(aq)$.
- (iii) In a second trial, using a different 25.00 mL sample of NH_2OH solution, the volume of HCl used to reach the equivalence point is greater than in the first trial. The student incorrectly claims that the 0.5604 M stock solution of HCl was used in the second trial instead of the diluted solution. Explain why the student's claim is incorrect.

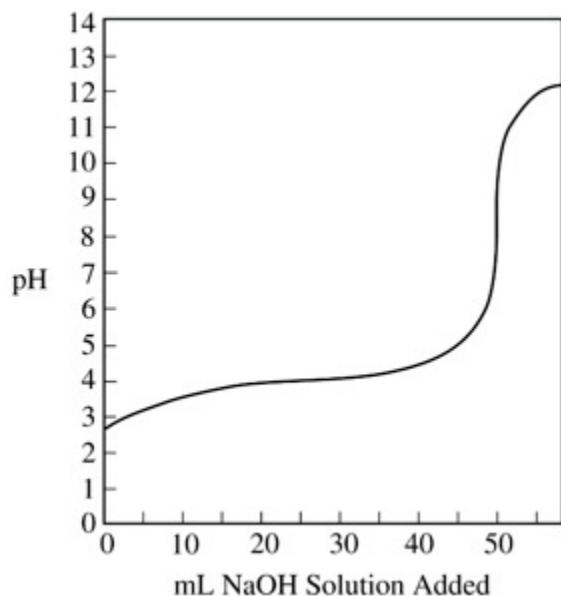


At elevated temperatures, $\text{NH}_2\text{OH}(aq)$ undergoes decomposition according to the equation shown.

- (f) If 8.00g of NH_2OH decomposes according to the equation above, how much heat, in kJ, is produced?

Chapter 11 Part 4: Titration

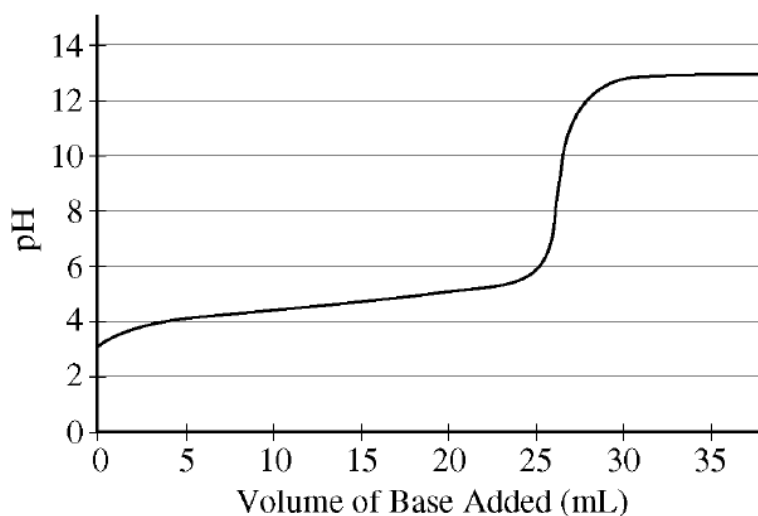
38.



The graph above shows the titration curve that resulted when a sample of 0.1 *M* monoprotic acid was titrated with a solution of NaOH. Based on the graph, the pK_a of the acid is closest to

- (A) 3.0
- (B) 4.0
- (C) 6.0
- (D) 8.0
- (E) 12.0

39.



A student performs an acid-base titration and plots the experimental results in the graph above. Which of the following statements best explains the experimental findings?

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- (A) A strong acid was titrated with a strong base, as evidenced by the equivalence point at $\text{pH} = 7$.
- (B) A strong acid was titrated with a strong base, as evidenced by the equivalence point at $\text{pH} > 7$.
- (C) A weak acid was titrated with a strong base, as evidenced by the equivalence point at $\text{pH} > 7$.
- (D) A weak acid was titrated with a weak base, as evidenced by the equivalence point at pH approximately 7.
-

Refer to the following.

Concentration (<i>M</i>)	pH of Acid 1	pH of Acid 2	pH of Acid 3	pH of Acid 4
0.010	3.44	2.00	2.92	2.20
0.050	3.09	1.30	2.58	1.73
0.10	2.94	1.00	2.42	1.55
0.50	2.69	0.30	2.08	1.16
1.00	2.44	0.00	1.92	0.98

The pH of solutions of four acids prepared at various concentrations were measured and recorded in the table above.

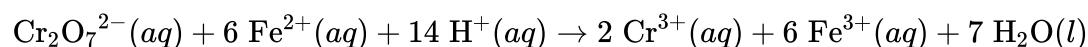
The four acids are, in no particular order, chlorous, hydrochloric, lactic, and propanoic.

40. If equal volumes of the four acids at a concentration of 0.50 *M* are each titrated with a strong base, which will require the greatest volume of base to reach the equivalence point?
- (A) Acid 1
- (B) Acid 2
- (C) Acid 3
- (D) All the acids will require the same volume of base to reach the equivalence point.
-

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41. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student determines the mass percent of iron in a sample of iron ore by doing a redox titration. The student finds the mass of the sample and then dissolves it in acid to a total volume of 25.00 mL. Using a buret, the student titrates the 25.00 mL solution with 0.017 M $\text{K}_2\text{Cr}_2\text{O}_7$. The following reaction occurs during the titration.



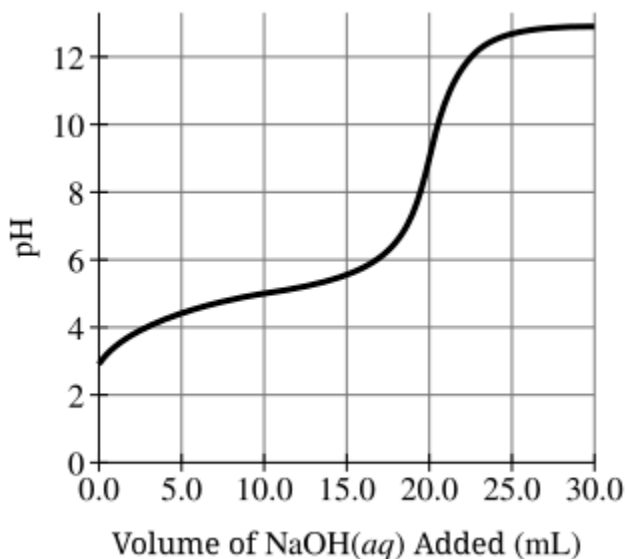
The student collects the following data.

Mass of iron ore in original sample	0.36 g
Buret reading at end point	45.52 mL
Initial buret reading	15.05 mL

- (a) How many moles of $\text{Fe}^{2+}(\text{aq})$ reacted with the $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$?
- (b) Assuming that all the iron in the ore was Fe^{2+} , calculate the mass percent of iron in the ore sample.
- (c) The student learns that the actual mass percent of iron in the ore sample is greater than the experimental value calculated in part (b). The student claims that the difference between the actual and experimental values could have been caused by rinsing the buret with distilled water just prior to filling it with 0.017 M $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$. Do you agree with the student's claim? Explain your reasoning.

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42.



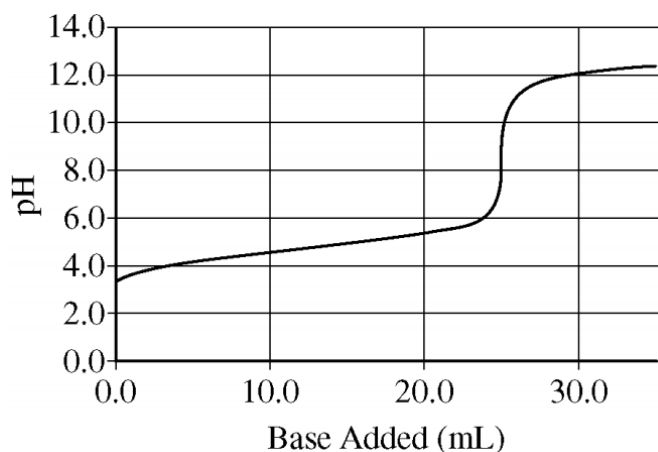
A 0.10 *M* solution of a weak acid is titrated with 0.10 *M* NaOH(*aq*). Based on the titration curve above, which of the following could be the identity of the weak acid?

- (A) HCN with a K_a value of 5×10^{-10}
 - (B) HOCl with a K_a value of 3×10^{-8}
 - (C) $\text{CH}_3\text{CH}_2\text{COOH}$ with a K_a value of 1×10^{-5}
 - (D) ClCH_2COOH with a K_a value of 1×10^{-3}
43. When a buret is rinsed before a titration, which of the techniques below is the best procedure?
- (A) Rinse the buret one time with some of the titrant solution.
 - (B) Rinse the buret one time with some of the titrant solution and then dry the buret in an oven.
 - (C) Rinse the buret two times: once with some of the titrant solution, then once with distilled water.
 - (D) Rinse the buret two times: each time with some of the titrant solution.
 - (E) Rinse the buret two times: each time with distilled water.

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A 0.35 g sample of $\text{Li}(s)$ is placed in an Erlenmeyer flask containing 100 mL of water at 25°C . A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the $\text{Li}(s)$ has reacted with $\text{H}_2\text{O}(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below



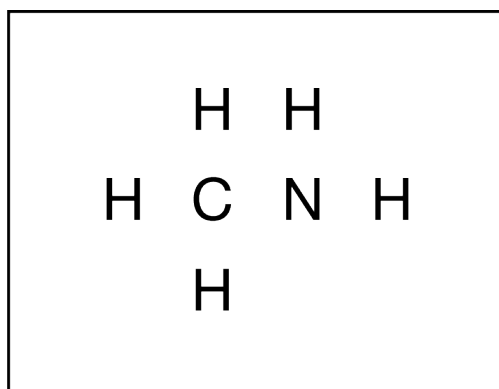
44. On the basis of the pH curve, the pK_a value of the acid is closest to
- (A) 4
 - (B) 5
 - (C) 8
 - (D) 12
-

Chapter 11 Part 4: Titration

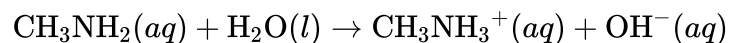
45. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Methylamine is a weak base with the formula CH_3NH_2 .

(a) In the following box, complete the Lewis electron-dot diagram for a molecule of methylamine. Show all bonding and nonbonding valence electrons.



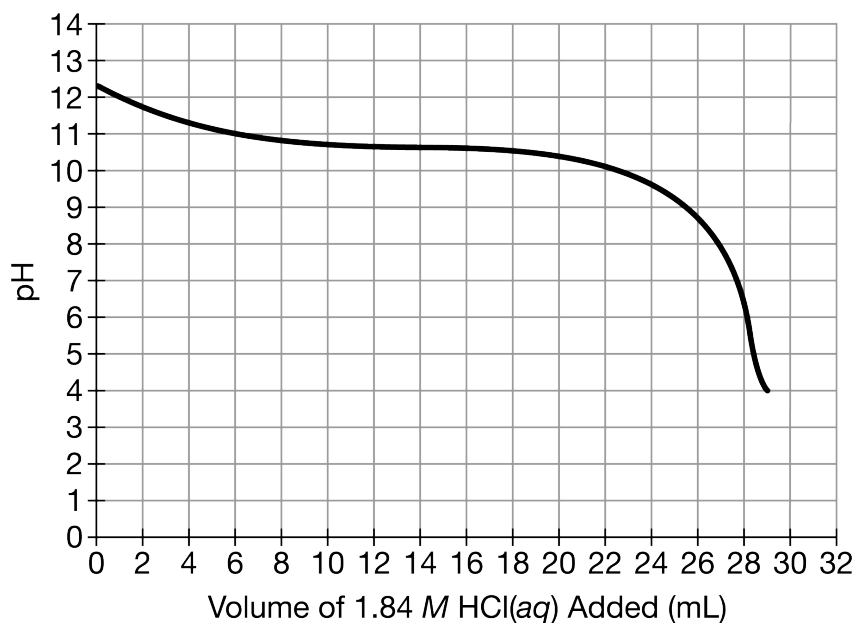
The following chemical equation represents the reaction that occurs when methylamine dissolves in water to form a basic solution.



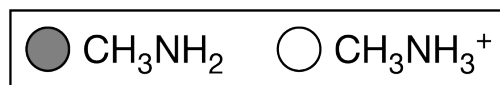
(b) The pH of 2.65 M $\text{CH}_3\text{NH}_2(aq)$ is 12.54. Determine the value of K_b for methylamine.

A 25.00 mL sample of a $\text{CH}_3\text{NH}_2(aq)$ solution of unknown concentration is titrated with 1.84 M $\text{HCl}(aq)$. Following is a graph that shows pH versus the volume of 1.84 M $\text{HCl}(aq)$ added during the titration.

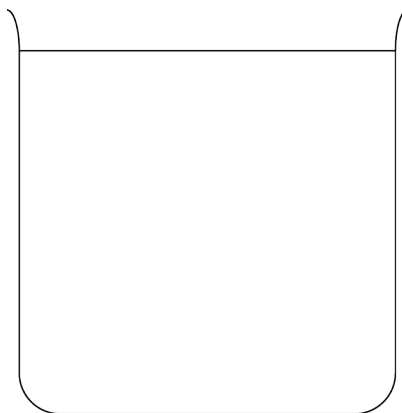
Chapter 11 Part 4: Titration



(c) If 28.25 mL of 1.84 M HCl(aq) was required to reach the equivalence point, calculate the concentration of the $\text{CH}_3\text{NH}_2(\text{aq})$ solution of unknown concentration.



(d) Using the symbols in the legend above, draw particles in the following beaker to represent the relative amounts of the two species, CH_3NH_2 and CH_3NH_3^+ , in the solution after the first 5.00 mL of titrant had been added to the $\text{CH}_3\text{NH}_2(\text{aq})$ solution.

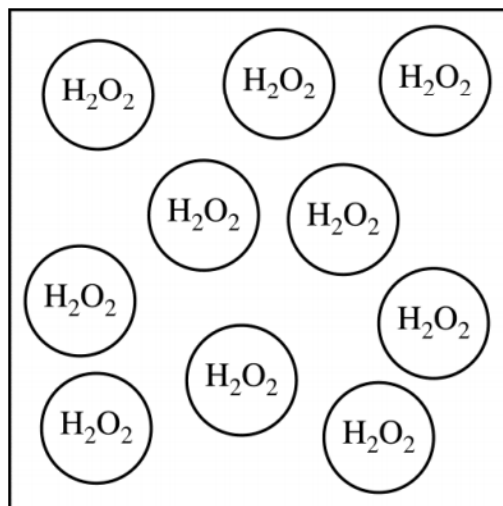


Beaker after 5.00 mL of
1.84 M HCl(aq) Added

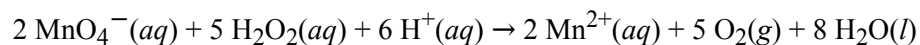
(e) Explain why the titration curve shows only a small change in pH per volume of acid added when the total amount of acid added is about 14.0 mL. Include a balanced chemical equation as part of your answer.

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46.



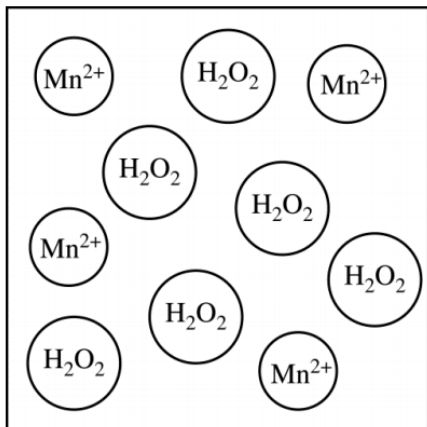
A particle view of a sample of $\text{H}_2\text{O}_2(aq)$ is shown above. The $\text{H}_2\text{O}_2(aq)$ is titrated with $\text{KMnO}_4(aq)$, as represented by the equation below.



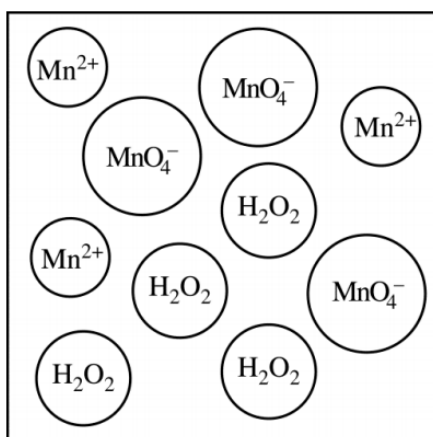
Which of the following particle views best represents the mixture when the titration is halfway to the equivalence point? (H_2O molecules and H^+ ions are not shown.)

Chapter 11 Part 4: Titration

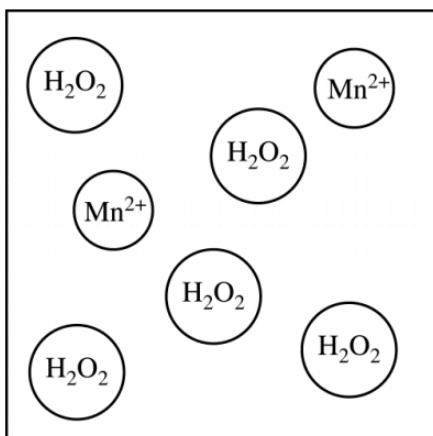
(A)



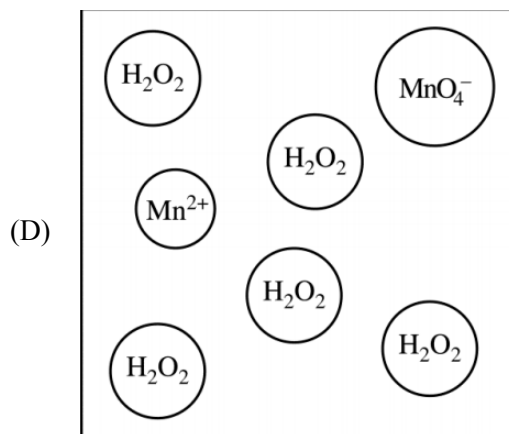
(B)



(C)



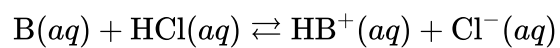
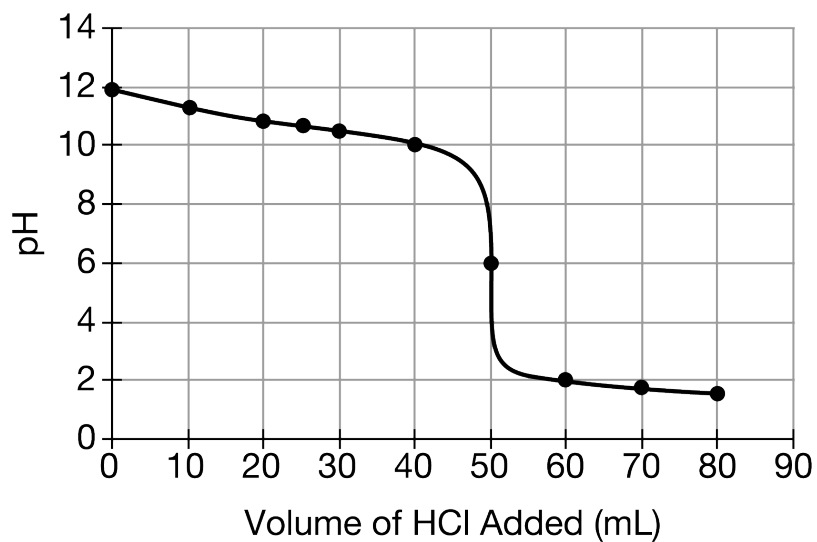
Chapter 11 Part 4: Titration



47. A 0.08 M solution of CH_3COOH ($\text{p}K_a = 4.74$) is titrated with $0.10\text{ M NaOH}(aq)$. What is the pH at the equivalence point of the titration and why?
- (A) $\text{pH} < 7$, because $\text{NaOH}(aq)$ is a strong base.
 - (B) $\text{pH} = 7$, because the titration reaction is a neutralization reaction.
 - (C) $\text{pH} > 7$, because $\text{CH}_3\text{COO}^-(aq)$ is a weak base.
 - (D) $\text{pH} > 7$, because the concentration of $\text{NaOH}(aq)$ is greater than that of $\text{CH}_3\text{COOH}(aq)$.
48. When a strong acid is titrated with a strong base using phenolphthalein as an indicator, the color changes abruptly at the endpoint of the titration and can be switched back and forth by the addition of only one drop of acid or base. The reason for the abruptness of this color change is that
- (A) a large change in pH occurs near the endpoint of the titration
 - (B) a buffer solution exists at the endpoint of the titration
 - (C) phenolphthalein is a strong proton donor
 - (D) the pH of water is very resistant to change
 - (E) phenolphthalein is much more sensitive to the pH of a solution than most other indicators

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49.



The graph above shows the titration curve for an aqueous solution of a weak base, B, with HCl as the titrant. Based on the graph, which of the following best estimates the $\text{p}K_a$ of HB^+ ?

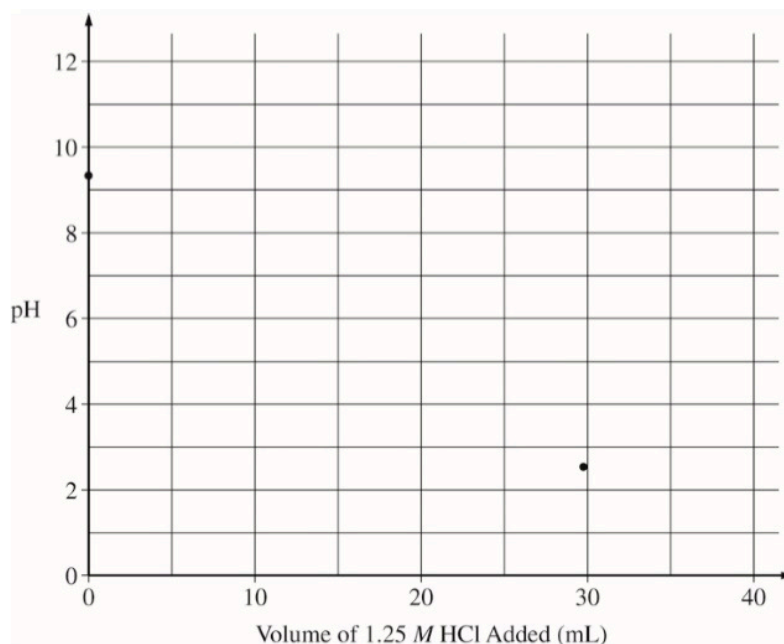
- (A) 12.0
- (B) 10.8
- (C) 6.0
- (D) 1.8

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50. Potassium sorbate, $\text{KC}_6\text{H}_7\text{O}_2$ (molar mass 150. g/mol) is commonly added to diet soft drinks as a preservative. A stock solution of $\text{KC}_6\text{H}_7\text{O}_2(aq)$ of known concentration must be prepared. A student titrates 45.00 mL of the stock solution with 1.25 M $\text{HCl}(aq)$ using both an indicator and a pH meter. The value of K_a for sorbic acid, $\text{C}_6\text{H}_7\text{O}_2$, is 1.7×10^{-5} .
- Write the net-ionic equation for the reaction between $\text{KC}_6\text{H}_7\text{O}_2(aq)$ and $\text{HCl}(aq)$.
 - A total of 29.95 mL of 1.25 M $\text{HCl}(aq)$ is required to reach the equivalence point. Calculate $[\text{KC}_6\text{H}_7\text{O}_2]$ in the stock solution.
 - The pH at the equivalence point of the titration is measured to be 2.54. Which of the following indicators would be the best choice for determining the end point of the titration? Justify your answer.

Indicator	$\text{p}K_a$
Phenolphthalein	9.3
Bromothymol blue	7.0
Methyl red	5.0
Thymol blue	2.0
Methyl violet	0.80

- Calculate the pH at the half-equivalence point.
- The initial pH and the equivalence point are plotted on the graph below. Accurately sketch the titration curve on the graph below. Mark the position of the half-equivalence point on the curve with an X.

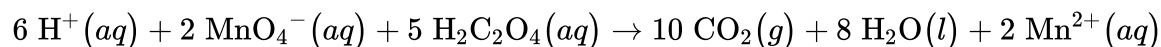


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- f. The pH of the soft drink is 3.37 after the addition of the $\text{KC}_6\text{H}_7\text{O}_2(aq)$. Which species, $\text{HC}_6\text{H}_7\text{O}_2$ or $\text{C}_6\text{H}_7\text{O}_2^-$, has a higher concentration in the soft drink? Justify your answer.

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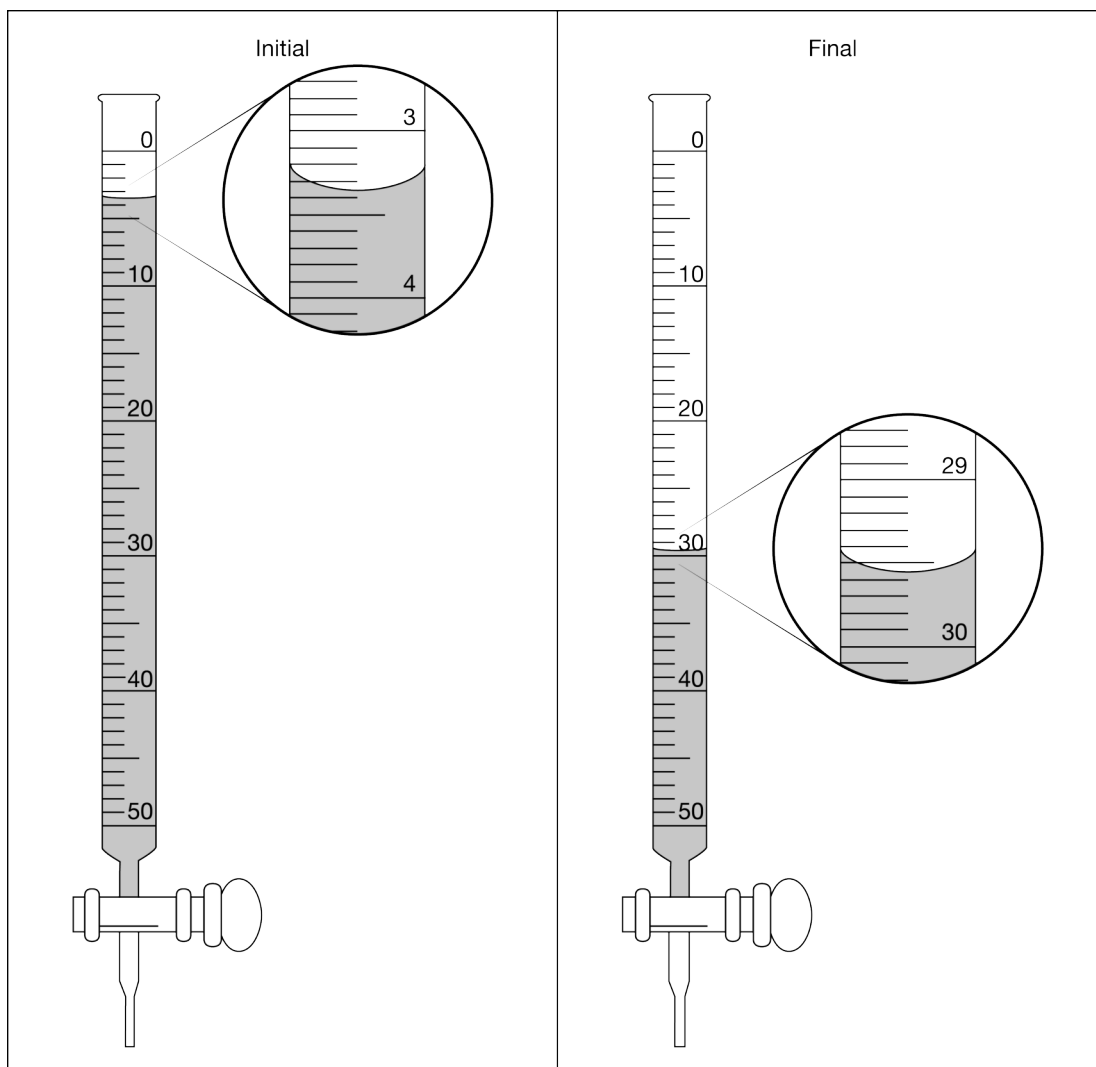
51. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student dissolved a 0.139 g sample of oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$, in water in an Erlenmeyer flask. Then the student titrated the $\text{H}_2\text{C}_2\text{O}_4$ solution in the flask with a solution of KMnO_4 , which has a dark purple color. The balanced chemical equation for the reaction that occurred during the titration is shown above.

- (a) Identify the species that was reduced in the titration reaction. Justify your answer in terms of oxidation numbers.
- (b) The student used a 50.0 mL buret to add the $\text{KMnO}_4(aq)$ to the $\text{H}_2\text{C}_2\text{O}_4(aq)$ until a faint lavender color was observed in the flask, an indication that the end point of the titration had been reached. The initial and final volume readings of the solution in the buret are shown below. Write down the initial reading and the final reading and use them to determine the volume of $\text{KMnO}_4(aq)$ that was added during the titration.

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(c) Given that the concentration of $\text{KMnO}_4(aq)$ was 0.0235 M , calculate the number of moles of MnO_4^- ions that completely reacted with the $\text{H}_2\text{C}_2\text{O}_4$.

(d) The student proposes to perform another titration using a 0.139 g sample of $\text{H}_2\text{C}_2\text{O}_4$, but this time using $0.00143\text{ M KMnO}_4(aq)$ in the buret. Would this titrant concentration be a reasonable choice to use if the student followed the same procedure and used the same equipment as before? Justify your response.

Chapter 11 Part 4: Titration

52. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student reacts 0.300 g of methyl salicylate ($\text{C}_8\text{H}_8\text{O}_3$) with a stoichiometric amount of a strong base. This product is then acidified to produce salicylic acid crystals ($\text{HC}_7\text{H}_5\text{O}_3$).

(a) For every 1 mole of $\text{C}_8\text{H}_8\text{O}_3$ (molar mass 152.15 g/mol) reactant used, 1 mole of salicylic acid crystals ($\text{HC}_7\text{H}_5\text{O}_3$, molar mass 138.12 g/mol) is produced. Calculate the maximum mass, in grams, of $\text{HC}_7\text{H}_5\text{O}_3$ that could be produced in this reaction.

As part of the experimental procedure to purify the $\text{HC}_7\text{H}_5\text{O}_3$ crystals after the reaction is complete, the crystals are filtered from the reaction mixture, rinsed with distilled water, and dried. Some physical properties of $\text{HC}_7\text{H}_5\text{O}_3$ are given in the following table.

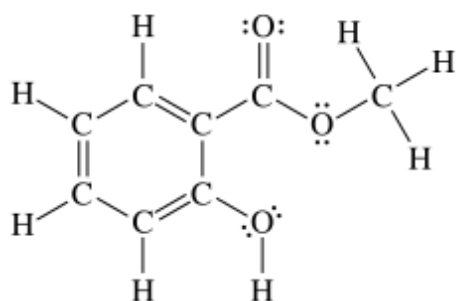
Properties of Salicylic Acid ($\text{HC}_7\text{H}_5\text{O}_3$)	
Melting point	159 °C
Solubility in H_2O at 25 °C	2.2 g/L
Specific heat capacity	1.17 J/(g·°C)
Heat of fusion	27.1 kJ/mol

(b) The student's experiment results in an 87% yield of dry $\text{HC}_7\text{H}_5\text{O}_3$. The student suggests that some of the $\text{HC}_7\text{H}_5\text{O}_3$ crystals dissolved in the distilled water during the rinsing step. Is the student's claim consistent with the calculated percent yield value? Justify your answer.

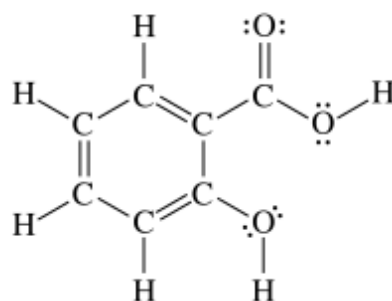
(c) Given the physical properties in the table, calculate the quantity of heat that must be absorbed to increase the temperature of a 0.105 g sample of dry $\text{HC}_7\text{H}_5\text{O}_3$ (molar mass 138.12 g/mol) crystals from 25 °C to the melting point of 159 °C and melt the crystals completely.

The structures and melting points for methyl salicylate and salicylic acid are shown.

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Methyl Salicylate
Melting Point: -9°C

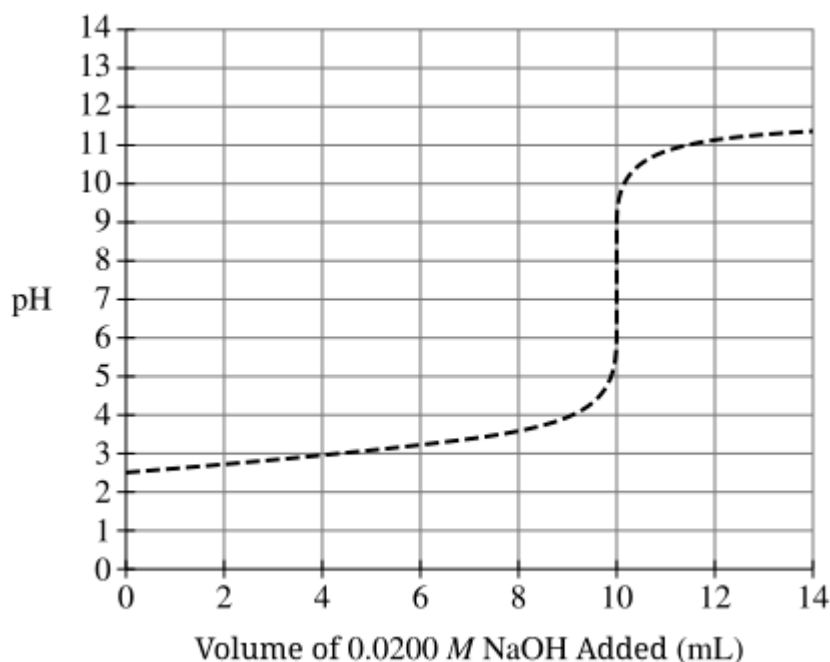


Salicylic Acid
Melting Point: 159°C

(d) The same three types of intermolecular forces (London dispersion forces, dipole-dipole interactions, and hydrogen bonding) exist among molecules of each substance. Explain why the melting point of salicylic acid is higher than that of methyl salicylate.

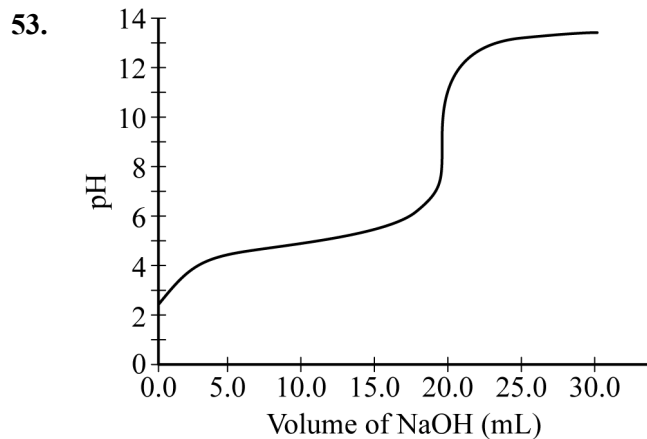
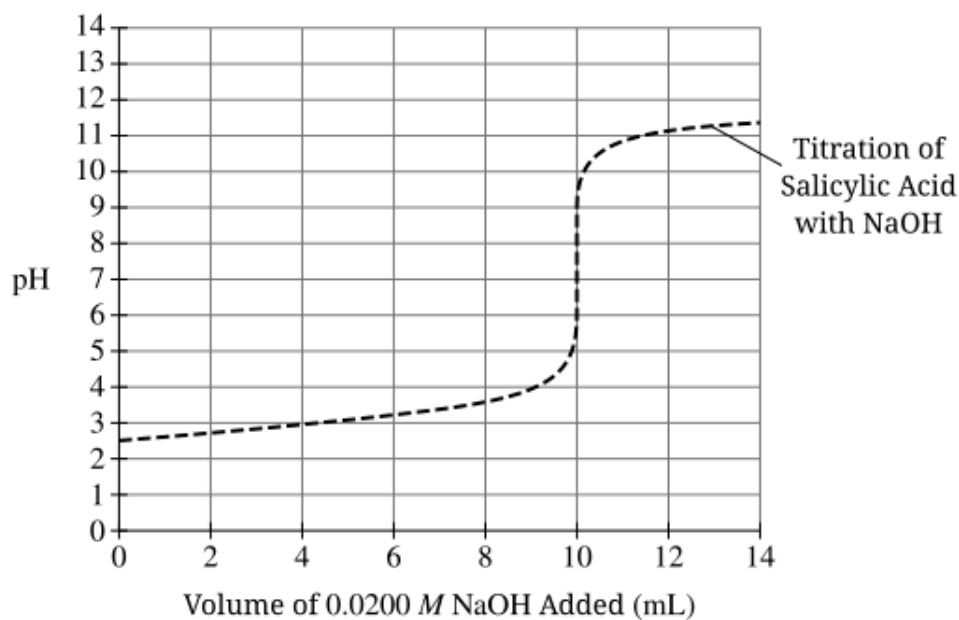
The student titrates 20.0 mL of 0.0100M $\text{HC}_7\text{H}_5\text{O}_3(aq)$ with 0.0200M NaOH, using a probe to monitor the pH of the solution. The data are plotted producing the following titration curve.

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- (e) Using the information in the graph, estimate the pK_a of $HC_7H_5O_3$. _____
- (f) When the pH of the titration mixture is 4.00, is there a higher concentration of the weak acid, $HC_7H_5O_3$, or its conjugate base, $C_7H_5O_3^-$, in the flask? Justify your answer.
- (g) The student researches benzoic acid ($HC_7H_5O_2$) and finds that it has similar properties to salicylic acid ($HC_7H_5O_3$). The K_a for benzoic acid is 6.3×10^{-5} . Calculate the value of pK_a for benzoic acid.
- (h) The student performs a second titration, this time titrating 20.0 mL of a 0.0100 M benzoic acid solution with 0.0200 M NaOH. Sketch the curve that would result from this titration of benzoic acid on the following graph, which already shows the original curve from the titration of 20.0 mL of 0.0100 M salicylic acid. The initial pH of the benzoic acid solution is 3.11.

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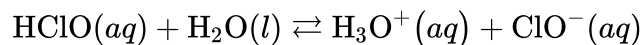
Data collected during the titration of a 20.0 mL sample of a 0.10 M solution of a monoprotic acid with a solution of NaOH of unknown concentration are plotted in the graph above. Based on the data, which of the following are the approximate pK_a of the acid and the molar concentration of the NaOH?

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(A)	pK_a	[NaOH]
	4.7	0.050 <i>M</i>
(B)	pK_a	[NaOH]
	4.7	0.10 <i>M</i>
(C)	pK_a	[NaOH]
	9.3	0.050 <i>M</i>
(D)	pK_a	[NaOH]
	9.3	0.10 <i>M</i>

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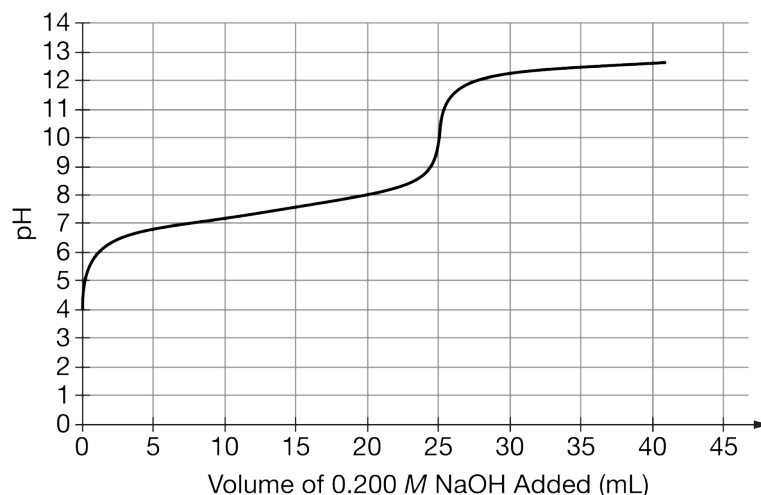
54. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The reaction between hypochlorous acid and water is represented above.

(a) Identify one of the conjugate acid-base pairs in the reaction.

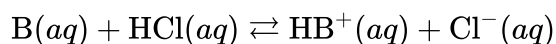
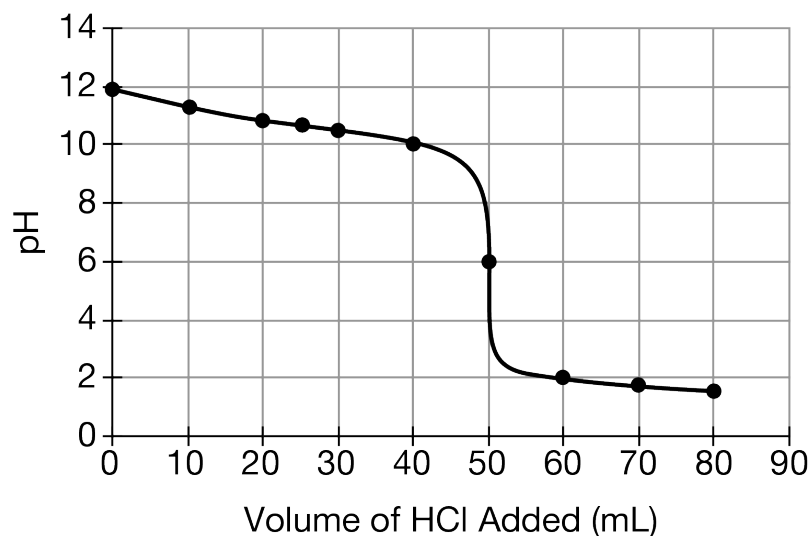
(b) Shown in the graph below is the titration curve that results when 25.0 mL of 0.200 *M* HClO is titrated with 0.200 *M* NaOH. Carefully draw a second curve on the graph that would result from the titration of 25.0 mL of 0.200 *M* HClO₄ with 0.200 *M* NaOH.



(c) A student proposes creating a buffer by dissolving 0.010 mol of NaClO₄(*s*) in 100. mL of 0.100 *M* HClO₄. Explain why the resulting solution would not be a buffer.

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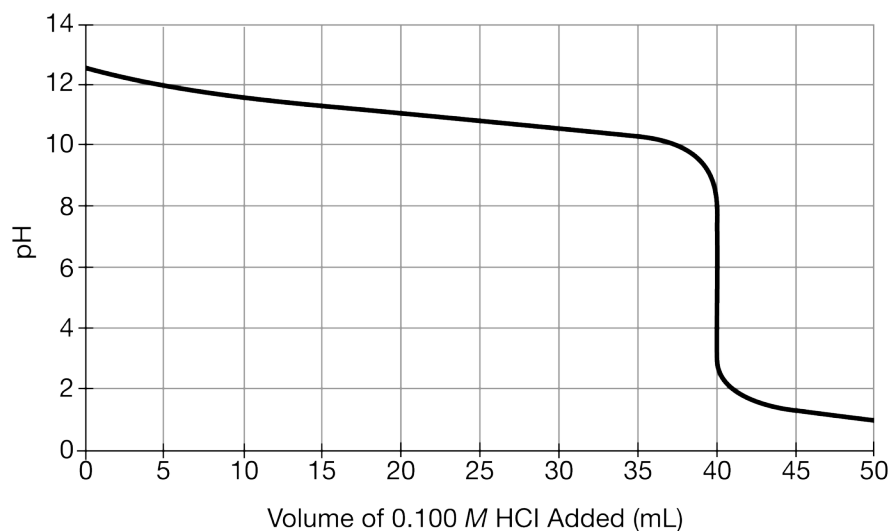
55.



The reaction of a weak base, B, with HCl is represented by the equation above. The graph shows the titration curve for 25.0 mL of an aqueous solution of B titrated with 0.100 M HCl. Based on the graph, which of the following best estimates the initial concentration of the solution of the weak base?

- (A) 0.20 M
- (B) 0.10 M
- (C) 0.067 M
- (D) 0.048 M

56.



A 40.0 mL sample of the weak base $\text{C}_5\text{H}_{11}\text{N}$ was titrated with 1.00 M HCl at 25 °C. Based on the resulting titration curve shown above, which of the following pairs provide the best estimates for the $\text{p}K_b$ and K_b of $\text{C}_5\text{H}_{11}\text{N}$?

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- (A) $\text{p}K_b \approx 3.0$ and $K_b \approx 1 \times 10^{-3}$
- (B) $\text{p}K_b \approx 5.8$ and $K_b \approx 2 \times 10^{-6}$
- (C) $\text{p}K_b \approx 11.0$ and $K_b \approx 1 \times 10^{-11}$
- (D) $\text{p}K_b \approx 12.5$ and $K_b \approx 3 \times 10^{-2}$